KAUSTUBH DWIVEDI

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PERSONAL STATEMENT

Mechanical and aerospace structures engineer with an MSc in Advanced Lightweight and Composite Structures from Cranfield University and a 78.83% weighted average. Combines finite element analysis, composite testing and manufacturing assessment to inform practical structural decisions. Delivered an aircraft component redesign with 20.6% lower mass and completed an engineering placement at Dassault Systemes. Contributed to **five peer-reviewed publications**, including first-author research on 3D-printed lattice mechanics. Experienced in coordinating project contributions, communicating technical findings and documenting verification evidence. Seeking a UK structures, design or analysis role.

EDUCATION

MSc, Advanced Lightweight and Composite Structures, Cranfield University, Cranfield, UK — **September 2025 - September 2026**

- **Weighted average: 78.83%** | Relevant study: composite structures, finite element analysis, structural stability and impact mechanics
- **Thesis:** Slat Track Cans of the Future: Exploring the Design Space of a High Criticality Fuel-Tank Component

B.Tech, Mechanical Engineering, Symbiosis Institute of Technology, Pune, India — **July 2020 - July 2024**

- CGPA: 8.36/10

PROJECTS

MSc Thesis: Slat Track Cans of the Future: Exploring the Design Space of a High Criticality Fuel-Tank Component

Cranfield University / GKN Aerospace, Cranfield, UK — **May 2026 - September 2026**

Tools: CATIA V5, **Abaqus**, **Tosca Structure**, **nCode** and **ANSYS Additive**.

- Redesigned an Airbus A320 slat-track can (fuel-tank component) in AlSi10Mg, **reducing mass by 20.6% to 1.478 kg** while retaining elastic response across five load cases and a minimum predicted yield factor of safety of 5.56.
- Devised a variable-wall design in a separate 1.450 kg comparison, reducing simulated displacement by 3.4-5.1% and stress by 3.7-10.1% while **avoiding local yielding** observed in the comparable uniform-wall model.
- **Predicted 1.145 million fatigue repeats** for the existing welded assembly against a 300,000-repeat target using **nCode** and EN 1999-1-3 curves; verified calculations with a separate method within 3%.
- Delivered a watertight **CATIA V5** solid and STEP model, verified geometry on an independent 239,329-element mesh and traced thermal hot spots to build direction through **ANSYS Additive** control simulations.
- Agreed five pressure cases with the industrial supervisor and screened four manufacturing routes, excluding two polymer options for low stiffness; documented remaining material-data, buckling and leak/proof-test requirements.

Team Lead: Composite Wing Leading-Edge Group Project

Cranfield University / GKN Aerospace, Cranfield, UK — **January 2026 - May 2026**

- **Led a four-person bird-strike project team**, allocating tasks, monitoring deadlines and integrating teammates' simulation and test inputs into design recommendations across five material systems.
- Ranked five leading-edge systems - CFRP, GFRP, Kevlar/epoxy, honeycomb sandwich and metal - at 90 and 180 m/s with **Abaqus/Explicit** SPH-FEA, comparing deformation, force response and energy absorption.
- Presented recommendations for five leading-edge material systems at design reviews, explaining laminate behaviour, manufacturing trade-offs, modelling assumptions and limitations to guide material selection.
- Reconciled **simulation and specimen-test results** with teammates in the four-person group, integrating analysis and experimental contributions into the bird-strike assessment.

CAREER HISTORY

Mohan Enviro Engineering Pvt Ltd, Noida, India, Mechanical Design Intern **December 2023 - April 2024**
Five-month on-site internship in water-treatment engineering

- Monitored and recorded **water-quality parameters daily** to support environmental compliance monitoring during a five-month internship.
- Analysed treatment-process data to evaluate system performance and maintained daily logs of operations, test results and maintenance activities.
- Partnered with engineers to create CAD models and diagrams for water-treatment system components.

La Fondation Dassault Systemes / Dassault Systemes, Pune, India, **December 2022 - December 2023**
Engineering Design Intern
Engineering design placement

- Developed a modular multi-ratio gearbox **from concept to working prototype** using **CATIA V5**, **3DEXPERIENCE**, analytical calculations and **Abaqus**.
- Refined modular gearbox geometry against three design constraints: force transmission, packaging and manufacturability, updating technical documentation after each iteration.
- Collaborated with senior engineers during the placement, translating formal review feedback into gearbox changes and maintaining technical documentation through design sign-off.

SKILLS, INTERESTS & EXTRACURRICULAR ACTIVITIES

- **Software: CATIA V5**, **3DEXPERIENCE**, **Abaqus/Standard**, **Abaqus/Explicit**, **ANSYS Additive**, **nCode**, **Tosca Structure**, **Python**, **MATLAB** and **Excel**.
- **Structural Engineering:** nonlinear finite element analysis, contact modelling, SPH impact, mesh sensitivity and independent-model verification.
- **Composites and Testing:** laminate and sandwich design, stacking sequences, specimen manufacture, polymer compression testing and experimental data interpretation.
- **Fatigue and Verification:** EN 1999-1-3 curves, CS 25.963 and CS 25.561 load definitions, welded-joint fatigue assessment and requirements verification matrices.
- **Data Analysis:** Applied particle swarm optimisation to two predictive models, Random Forest and XGBoost, to estimate lattice mechanical performance from experimental manufacturing parameters.
- **Additive Manufacturing:** wall-thickness design, build orientation, support strategy and manufacturing-risk assessment.
- **Professional membership:** Royal Aeronautical Society, Student Affiliate, since November 2025.

PUBLICATIONS

- Designed, manufactured and compression-tested PLA+ diamond lattices to link printing parameters and cell geometry with mechanical performance, producing **first-author research** in Materials Today Communications, DOI 10.1016/j.mtcomm.2024.108233 (2024).
- Collaborated with co-authors on a published comparison of five aerospace material systems under bird-strike loading in the Journal of Manufacturing and Materials Processing, DOI 10.3390/jmmp10090343 (Cranfield University, 2026).
- Co-authored three peer-reviewed chapters covering polymer weld-strength optimisation and classification of dimensional deviation and surface roughness in 3D-printed parts, DOI 10.1201/9781003437369 (CRC Press, 2024).